

Marwan Badr

Software Developer | Backend, ERP & AI Systems

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Portfolio

PROFESSIONAL SUMMARY

Software Developer specializing in robust backend architectures, ERP engineering (Frappe/ERPNext, Odoo), and Agentic AI workflows. Combines an accounting background (B.Com in Accounting, English Section) with production experience in Python, FastAPI, and LangGraph to build high-performance APIs, custom enterprise business modules, and autonomous multi-agent LLM systems. Dedicated to clean architecture, financial data integrity, and end-to-end business automation.

TECHNICAL SKILLS

AI & Agentic Systems: LangGraph, LangChain, Multi-Agent Orchestration, Tool Calling, RAG, Structured Output
ERP Frameworks: Frappe Framework, ERPNext (v14 & v15), Odoo Development (v17 & v18), DocTypes, Hooks, Server Scripts
Backend & APIs: Python, FastAPI, Django, Django REST Framework (DRF), RESTful APIs, Webhooks, SSE Streaming
Databases & Cache: PostgreSQL, MariaDB, MySQL, Redis (caching & task state), Qdrant (Vector DB), SQLAlchemy
Languages & Web: Python, JavaScript, TypeScript, HTML5, CSS3, Tailwind CSS, Next.js, React, PyQt5
DevOps & Tools: Docker, Git, GitHub, Linux, Uvicorn, Caddy, CI/CD Workflows
Accounting Domain: Professional Financial Accounting (PFA), General Ledgers, Bank Reconciliation, Periodic Audit, Chart of Accounts

PROFESSIONAL EXPERIENCE

Showsatellite — Software Developer Mar 2026 – Present | Remote (Bahrain)

Premier Home Appliances Retail Company in Bahrain

- Lead custom Frappe / ERPNext engineering: custom DocTypes, client/server scripts, and automated accounting workflows.
- Engineered high-throughput real-time sync engine between ERPNext and OpenCart automating products, inventory, prices, and media.
- Built centralized operations logistics dashboard managing TV/AC delivery scheduling, real-time driver dispatch, and support ticketing.

Minicash Furniture Company — Accountant Feb 2026 – Mar 2026 | Onsite (Egypt)

- Managed daily accounting operations, general ledger journal entries, AR/AP reconciliations, and trial balance verification.

FEATURED PROJECTS

Razyyn AI — Autonomous AI Accountant Platform (Founder & Lead Architect) Python • FastAPI • LangGraph • PostgreSQL • Redis • Qdrant

- Architected an autonomous AI virtual finance SaaS integrating natively with ERPNext and Odoo (razyyn.com).
- Engineered Master Router coordinating 6 specialist desks (Ask, Analyse, Reconcile, Audit, Record, Update) with LangGraph & FastAPI.
- Built 10-tier algorithmic bank reconciliation engine matching statements to ledgers with zero unexplained discrepancies.
- Implemented dual-gated write security (ERP RBAC + Razyyn Write Policy with caps & blocked accounts) with human approval cards.
- Developed AST SQL query rewriting engine (`sqlglot`) enforcing live/posted rows across all queries, isolating draft/cancelled records.

Razyyn AI — Native Odoo Integration Modules (v17 & v18) Odoo 17 • Odoo 18 • Python • XML • JavaScript • PostgreSQL • SSE

- Built custom native Odoo addons for Odoo 17 & 18 (self-hosted & Odoo.sh) embedding the AI assistant into the Odoo web client.
- Implemented in-ERP chat UI with SSE streaming, conversation persistence, and interactive approval cards for draft invoices & entries.
- Created guarded HTTP write gateway with preflight schema validation, draft invoice creation, and immutable audit logs.
- Built safe model browsing and AST live row filters ensuring read isolation and preventing Cartesian joins across Odoo models.

Razyyn AI — Native Frappe Applications (v14 & v15) Frappe Framework (v14 & v15) • ERPNext • Python • JavaScript • MariaDB • JWT

- Developed custom Frappe apps for Frappe v14 & v15 providing an in-ERP desk chat page with streaming and Mermaid diagram rendering.
- Implemented granular Agent Write Policy DocTypes (amount caps, blocked accounts, allowed DocTypes) preventing unauthorized writes.
- Built financial attachment processor and OCR pipeline handling PDF, Excel (XLSX), CSV, Word, and receipts.

Integrated E-Commerce & Enterprise Operations Platform Python • FastAPI • ERPNext • PostgreSQL • REST APIs

- Built full retail platform integrated with ERPNext automating Sales Orders, Delivery Notes, and driver logistics ([Live Site](#)).

Real-Time Synchronization Engine: ERPNext ↔ OpenCart Python • FastAPI • REST APIs • ERPNext • OpenCart

- Engineered bidirectional sync engine automating product catalogs, price tiers, stock, and images ([Demo Video](#)).

Office Leaks — Professional Corporate Social Network Python • Django • Django REST Framework (DRF) • PostgreSQL

- Corporate transparency platform with company reviews, workplace culture ratings, and secure DRF REST APIs ([GitHub](#)).

Telegram Notification Manager (Frappe App) Frappe Framework • Python • Telegram Bot API

- Hook-driven notification app triggering real-time interactive Telegram alerts with action buttons for Frappe DocTypes ([GitHub](#)).

EDUCATION & CERTIFICATIONS

Menofia University — Bachelor of Commerce, Accounting (English Section) — Grade: Very Good Graduated 2025

Certifications & Training: Professional Financial Accountant (PFA) • Frappe / ERPNext Development • Odoo Development & Functional Accounting • Web Development with FastAPI & SQLAlchemy • Prompt Engineering

ADDITIONAL INFORMATION

Languages: Arabic (Native), English (Professional Working Proficiency) | **Military Status:** Permanently Exempted